

(TR-104) PYTHON WITH AI/ML

Training Day 34 Report

Date: 26 February 2026

Topic Covered: Parameter Efficient Fine-Tuning (LoRA/PEFT) and Transformer Optimization

The thirty-fourth day of training focused on understanding the limitations of full fine-tuning in transformers and implementing Parameter Efficient Fine-Tuning (PEFT) using LoRA (Low Rank Adaptation). The work included optimizing transformer training, reducing trainable parameters, and comparing LoRA with full fine-tuning approaches.

- Understood why full fine-tuning of transformer models is computationally expensive and memory intensive.
- Learned the concept of Parameter Efficient Fine-Tuning (PEFT) for optimizing transformer training.
- Explored how LoRA (Low Rank Adaptation) reduces trainable parameters while preserving model performance.
- Understood the mathematical intuition behind LoRA using low-rank matrix decomposition.
- Installed and configured Hugging Face PEFT and Accelerate libraries.
- Loaded DistilBERT model for sequence classification tasks.
- Created tokenization pipeline and applied dataset preprocessing using Hugging Face datasets library.
- Configured LoRA parameters including rank (r), lora_alpha, target_modules, and lora_dropout.
- Attached LoRA adapters to the pretrained DistilBERT model.
- Observed the reduction in trainable parameters after applying LoRA.
- Performed training using Hugging Face Trainer API with LoRA-enabled model.
- Evaluated the model using Accuracy and F1-score metrics.
- Compared LoRA and Full Fine-Tuning in terms of training time, trainable parameters, model accuracy, checkpoint size, and memory efficiency.
- Understood when to use LoRA and when full fine-tuning is suitable.

- Learned how PEFT enables efficient multi-task and production-level transformer deployment.
- Strengthened understanding of transformer optimization and efficient training strategies for real-world NLP systems.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/Transformers/tree/main/peft>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/lora-peft-fine-tuning-transformers-3a7a9dd0f40c>